

Chapter 2 Seminar: Simple Linear Regression

Student version

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x	y
-1	1
0	0
1	3
2	2
3	4

Exercise 2.1

This exercise uses a very small dataset so that the main calculations behind ordinary least squares (OLS) can be followed by hand.

Parts **(a)**–**(j)** follow the structure of the original textbook exercise. Parts **(k)**–**(p)** are extensions used in the seminar to connect the calculations to OLS properties, variation, assumptions, R output, and extrapolation.

Each part is followed immediately by space for your own solution.

The five observations are:

```
exercise_data <- tibble(
  x = c(-1, 0, 1, 2, 3),
  y = c(1, 0, 3, 2, 4)
)

exercise_data |>
  gt() |>
  fmt_number(decimals = 0)
```

2.1(a) Sample means and deviations from the means

Calculate the sample means $\bar{x}$ and $\bar{y}$.

Then complete the table below. For each observation, calculate the deviation of x_i from its mean, $x_i - \bar{x}$, and the deviation of y_i from its mean, $y_i - \bar{y}$. Also calculate the squared deviations and the cross-product of the two deviations.

x_i	y_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
-1	1					
0	0					
1	3					
2	2					
3	4					
Sum						

Use the completed table to calculate the sums needed later in the exercise.

Also consider the following:

1. What does a positive or negative value of $x_i - \bar{x}$ mean?

2. What does a positive or negative value of $y_i - \bar{y}$ mean?
3. What do you notice about $\sum(x_i - \bar{x})$ and $\sum(y_i - \bar{y})$?
4. Plot the observations and show $\bar{x}$ and $\bar{y}$ on the graph.

Your solution

2.1(b) Estimate the regression line

Use the OLS slope formula

$$b_2 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

to calculate b_2 . Then calculate the intercept from $b_1 = \bar{y} - b_2\bar{x}$.

Write the fitted regression line as $\hat{y}_i = b_1 + b_2x_i$ and interpret both b_1 and b_2 .

Your solution

2.1(c) Alternative formulas for the OLS sums

Calculate $\sum x_i^2$ and $\sum x_i y_i$.

Use these values to verify the two equivalent formulas

$$\sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N x_i^2 - N\bar{x}^2$$

and

$$\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}.$$

The point of this part is to see that the same quantities can be calculated either from deviations around the means or from raw sums.

Your solution

2.1(d) Fitted values, residuals, and descriptive statistics

For each observation, calculate the fitted value $\hat{y}_i = b_1 + b_2 x_i$ and the residual $\hat{e}_i = y_i - \hat{y}_i$.

Complete the table:

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
-1	1				
0	0				
1	3				
2	2				
3	4				
Sum					

Then calculate the following by hand. The formulas are included here so that you can see exactly what each quantity measures.

- **Sample variance of x :** measures how spread out the observed x -values are around $\bar{x}$,

$$s_x^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}.$$
- **Sample variance of y :** measures how spread out the observed y -values are around $\bar{y}$,

$$s_y^2 = \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1}.$$

- **Sample covariance between x and y :** measures whether x and y tend to be above or below their means together, $s_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N-1}$.
- **Sample correlation:** standardizes the covariance so that the result lies between -1 and 1 , $r_{xy} = \frac{s_{xy}}{s_x s_y}$.
- **Coefficient of variation of x :** compares the standard deviation of x with its mean, $CV_x = 100 \frac{s_x}{\bar{x}}$.
- **Median of x :** the middle observation after the x -values have been sorted.

For each quantity, report the numerical result and briefly explain what it measures. After the manual calculations, use `var()`, `cov()`, `cor()`, and `median()` in R to verify the results.

Your solution

2.1(e) Plot the data, fitted values, and residuals

Plot the original observations and the fitted regression line.

On the same graph:

- show the fitted value $\hat{y}_i$ for each observation,
- show the residual $\hat{e}_i$ as the vertical distance between y_i and $\hat{y}_i$.

Use the graph to explain the difference between an observed value, a fitted value, and a residual.

Your solution

2.1(f) The point of sample means

On the graph from part (e), locate the point

$$(\bar{x}, \bar{y}).$$

Does the fitted regression line pass through this point?

Explain what you observe.

Your solution

2.1(g) Verify the mean-point property algebraically

Using the numerical values from this dataset, verify that

$$\bar{y} = b_1 + b_2 \bar{x}.$$

Explain why this result follows directly from the formula for b_1 .

Your solution

2.1(h) Mean of the fitted values

Calculate

$$\bar{\hat{y}} = \frac{1}{N} \sum_{i=1}^N \hat{y}_i.$$

Verify that

$$\bar{\hat{y}} = \bar{y}.$$

Use the relationship

$$y_i = \hat{y}_i + \hat{e}_i$$

to explain why this happens when the regression includes an intercept.

Your solution

2.1(i) Estimate the error variance

Calculate the sum of squared least squares residuals

$$SSE = \sum_{i=1}^N \hat{e}_i^2.$$

Then calculate the estimated error variance, which measures the remaining variation around the fitted regression line,

$$\hat{\sigma}^2 = \frac{SSE}{N-2}.$$

Also calculate

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2}.$$

Why is the denominator $N - 2$ rather than N ?

Draw a residual plot and use it to explain what the residual variation represents.

Your solution

2.1(j) Variance and standard error of the slope estimator

Let

$$S_{xx} = \sum_{i=1}^N (x_i - \bar{x})^2.$$

Calculate the estimated sampling variance of the slope,

$$\widehat{\text{Var}}(b_2 \mid \mathbf{x}) = \frac{\hat{\sigma}^2}{S_{xx}}$$

and

$$se(b_2) = \sqrt{\widehat{\text{Var}}(b_2 \mid \mathbf{x})}.$$

Explain the two parts of the formula:

- What happens to the precision of b_2 when the error variation becomes larger?
- What happens to the precision of b_2 when the observed x -values are more spread out?

Extensions

Your solution

Extensions

2.1(k) Variance and standard error of the intercept estimator

Calculate the estimated sampling variance of the intercept,

$$\widehat{\text{Var}}(b_1 \mid \mathbf{x}) = \hat{\sigma}^2 \left(\frac{1}{N} + \frac{\bar{x}^2}{S_{xx}} \right).$$

Show that this is equivalent to

$$\widehat{\text{Var}}(b_1 \mid \mathbf{x}) = \hat{\sigma}^2 \left[\frac{\sum_{i=1}^N x_i^2}{N S_{xx}} \right].$$

Then calculate

$$se(b_1) = \sqrt{\widehat{\text{Var}}(b_1 \mid \mathbf{x})}.$$

Your solution

2.1(l) Decomposing the variation in y

For each observation,

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{e}_i.$$

Use a graph to identify:

- the total deviation $y_i - \bar{y}$,
- the fitted component $\hat{y}_i - \bar{y}$,
- the residual component $\hat{e}_i$.

Then calculate

$$TSS = \sum_{i=1}^N (y_i - \bar{y})^2,$$

$$ESS = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2,$$

and

$$SSE = \sum_{i=1}^N \hat{e}_i^2.$$

Verify that

$$TSS = ESS + SSE.$$

Compare the sample variances

$$\text{Var}(y), \quad \text{Var}(\hat{y}), \quad \text{Var}(\hat{e}).$$

Finally calculate

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSE}{TSS}.$$

Interpret the result carefully. Does this decomposition imply causality?

Your solution

2.1(m) Why do we calculate $\sum x_i \hat{e}_i$?

Verify numerically that

$$\sum_{i=1}^N \hat{e}_i = 0$$

and

$$\sum_{i=1}^N x_i \hat{e}_i = 0.$$

Then explain why these two results arise from minimizing

$$SSE(b_1, b_2) = \sum_{i=1}^N (y_i - b_1 - b_2 x_i)^2.$$

What does

$$\sum_{i=1}^N x_i \hat{e}_i = 0$$

tell us about the sample covariance between x and the OLS residuals?

Does this prove that the population assumption

$$E(e_i \mid \mathbf{x}) = 0$$

is true? Explain.

Your solution

2.1(n) OLS properties versus model assumptions

For each statement below, decide whether it is:

- a **mechanical property of the fitted OLS regression**,
- a **model assumption about the population or data-generating process**, or
- a **sample condition that can be checked directly**.

Consider:

$$\sum_{i=1}^N \hat{e}_i = 0,$$

$$\sum_{i=1}^N x_i \hat{e}_i = 0,$$

$$\bar{\hat{y}} = \bar{y},$$

$$E(e_i \mid \mathbf{x}) = 0,$$

$$\text{Var}(e_i \mid \mathbf{x}) = \sigma^2,$$

$$\text{Cov}(e_i, e_j \mid \mathbf{x}) = 0 \quad i \neq j,$$

and the requirement that x takes at least two different values.

Briefly connect these statements to assumptions SR1–SR6.

Your solution

2.1(o) Verify the hand calculations using `lm()`

Only after completing the calculations above, estimate the model using

```
lm(y ~ x, data = exercise_data)
```

Use R to verify:

- b_1 ,
- b_2 ,
- the fitted values,
- the residuals,
- $\hat{\sigma}$,
- $se(b_1)$,
- $se(b_2)$.

Which quantities that took several steps to calculate by hand are produced automatically by `lm()`?

Your solution

2.1(p) Prediction and extrapolation

Use the fitted line to predict y for:

$$x = 2$$

and

$$x = 10.$$

The observed range of x in the sample is

$$-1 \leq x \leq 3.$$

R can calculate both predictions. Should we have the same confidence in both?

Plot the fitted line over an extended range and identify which prediction is an interpolation and which is an extrapolation.

Your solution

Exercise 2.2: Reproducing OLS with `lm()`

This exercise uses the six observations from textbook Exercise 2.3. The purpose is **not** to repeat all the hand calculations from Exercise 2.1. Instead, use R to reproduce the most important results with a few commands and connect the output back to what you calculated by hand.

Create the data:

```
exercise_22 <- tibble(  
  x = 1:6,  
  y = c(6, 4, 11, 9, 13, 17)  
)
```

Useful commands include `lm()`, `coef()`, `fitted()`, `residuals()`, `sigma()`, `vcov()`, `mean()`, and `sum()`.

2.2(a) Estimate the model

Estimate the simple linear regression

$$y_i = \beta_1 + \beta_2 x_i + e_i$$

using `lm()`.

Report the estimated regression line and interpret the intercept and slope.

Your solution

2.2(b) Plot the data and fitted line

Make a scatterplot of y against x and add the fitted regression line.

You can use `ggplot()`, `geom_point()`, and `geom_smooth(method = "lm", se = FALSE)`.

Add green dashed lines at $\bar{x}$ and $\bar{y}$. Does the fitted line pass through $(\bar{x}, \bar{y})$?

Your solution

2.2(c) Fitted values and residuals

Use `fitted()` and `residuals()` to obtain $\hat{y}_i$ and $\hat{e}_i$.

Create a table containing x_i , y_i , $\hat{y}_i$, and $\hat{e}_i$.

Choose one observation and explain the meaning of its fitted value and residual.

Your solution

2.2(d) Verify the OLS properties from Exercise 2.1

Use R to verify the following sample properties:

$$\sum_{i=1}^N \hat{e}_i = 0,$$

$$\sum_{i=1}^N x_i \hat{e}_i = 0,$$

and

$$\bar{\hat{y}} = \bar{y}.$$

Which of these are mechanical properties of OLS, and which are assumptions about the population?

Your solution

2.2(e) SSE and the estimated error variance

Use the residuals to calculate

$$SSE = \sum_{i=1}^N \hat{e}_i^2.$$

Then compare your calculation with `sigma(model)^2`.

What does `sigma(model)` report?

Your solution

2.2(f) Standard errors of the coefficients

Use R to obtain the standard errors of b_1 and b_2 .

You may use either

```
coef(summary(model))
```

or

```
vcov(model)
```

Explain how the standard error of b_2 relates to the calculation you carried out in Exercise 2.1.

For now, ignore the t-values and p-values in `summary(model)`. We return to statistical inference in Chapter 3.

Your solution

2.2(g) What did R replace?

For each item below, identify the R command that replaces the longer hand calculation from Exercise 2.1:

- b_1 and b_2
- fitted values
- residuals
- SSE
- $\hat{\sigma}$
- $se(b_1)$ and $se(b_2)$

What do you gain by doing Exercise 2.1 by hand before using `lm()`?

Your solution

Additional exercises if time allows

Exercise 2.3: Prediction and elasticity

This exercise is adapted from textbook Exercise 2.11.

Let y be monthly expenditure on food away from home per household member, measured in dollars, and let x be monthly household income measured in **hundreds of dollars**.

The estimated regression is

$$\hat{y} = 13.77 + 0.52x.$$

2.3(a) Interpret the coefficients

Interpret the estimated slope 0.52 carefully, remembering the units of x .

Also interpret the intercept. Is the intercept economically interesting in this application?

Your solution

2.3(b) Prediction

Predict monthly food-away-from-home expenditure for a household with monthly income of \$2000.

Remember that income is measured in hundreds of dollars.

Your solution

2.3(c) Elasticity at a point

For a linear regression, the elasticity at a particular value of x is

$$\varepsilon = b_2 \frac{x}{\bar{y}}.$$

Calculate the income elasticity of food-away-from-home expenditure at a monthly household income of \$2000.

Interpret the result.

Your solution

Exercise 2.4: Regression with an indicator variable

This exercise is adapted from textbook Exercise 2.12.

Let y be monthly food-away-from-home expenditure per household member. Define

- $ADVANCED = 1$ if the household includes a member with an advanced degree,
- $ADVANCED = 0$ otherwise.

The estimated regression is

$$\hat{y} = 44.96 + 30.41 ADVANCED.$$

2.4(a) Interpret the intercept and slope

What does the intercept 44.96 represent?

What does the coefficient 30.41 represent?

Your solution

2.4(b) Calculate the two group means

Use the regression equation to calculate predicted expenditure when $ADVANCED = 0$ and when $ADVANCED = 1$.

Show that the slope coefficient is exactly the difference between these two predicted group means.

Your solution